

*smoke along the walked polyline
sets what a route costs*

*sign legibility (fdsvismap)
sets what the agent knows*

Agent state

$t, \mathbf{x}_i, D_i$, current exit

FDS fields via fdsreader

$K(\mathbf{x}, t)$, gases, T · one slice at 2 m

Cognitive map

known nodes and edges
full | discovery

Restrict

to the subgraph this agent knows
(its cognitive map)

1 · Edge weights

$w_{uv} = \bar{K}_{uv} L_{uv} + \varepsilon L_{uv}$
optical depth per edge
length as tiebreak

2 · Dijkstra

one minimum-cost path
per reachable exit

3 · Route ranking

sort by $\tau_k = \bar{K}_k L_k$, then time R_k
congestion enters here only

every reevaluation, staggered across agents

Target exit

k^* and its path, handed to
JuPedSim as the journey

Switch rule

adopt a rival only if it beats the
current exit by a margin (10 %)

All refused?

reinstate the least- τ route
the agent always has a path

Feasibility

exposure gate $\tau_k > \mu_\tau \tau_{\max}$
dose veto $D_k^{\max} > D_{\text{thresh}}$